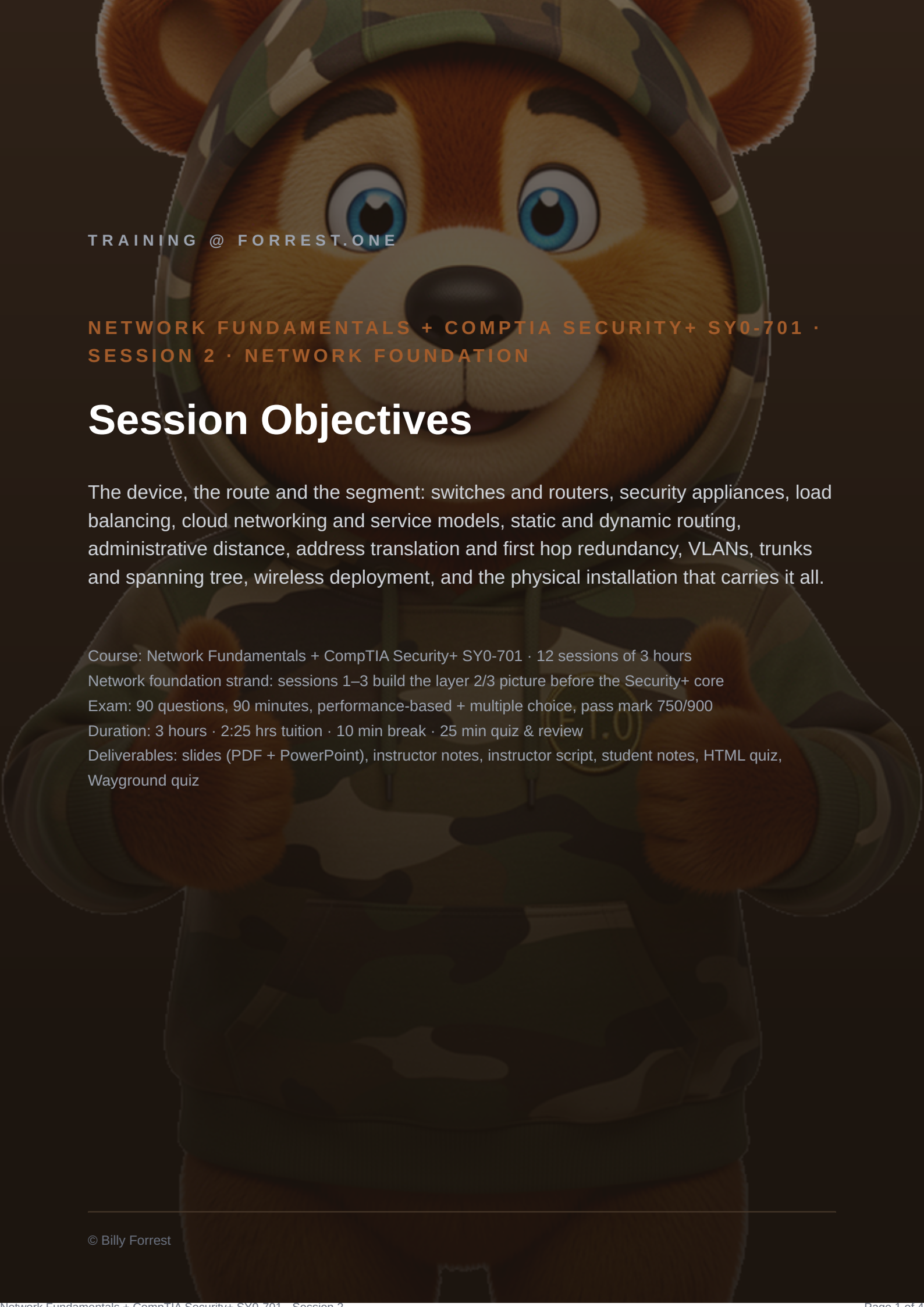




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 2 · NETWORK FOUNDATION**

Session Objectives

The device, the route and the segment: switches and routers, security appliances, load balancing, cloud networking and service models, static and dynamic routing, administrative distance, address translation and first hop redundancy, VLANs, trunks and spanning tree, wireless deployment, and the physical installation that carries it all.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Switching, VLANs & Network Architectures

This second session of the network-foundation strand moves from addressing to forwarding, and then outward to the devices, routes and segments that carry traffic. It covers the infrastructure devices on a modern network: switches, routers, firewalls, intrusion detection and prevention, load balancers, proxies, storage and wireless hardware. Then cloud networking and its service models, the routing protocols and route-selection rules that choose a path, address translation and first hop redundancy, VLANs, trunking and spanning tree, and finally wireless and physical deployment. Every mechanism is taught twice: once as plumbing, once as a boundary. Segmentation is a first-class Security+ control, because it reduces the attack surface, contains blast radius and blocks lateral movement, so the architecture material here carries directly into Domain 3 (Security Architecture) and Domain 4 (Operations). Sessions 4–12 assume it.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Compare a switching decision with a routing decision, and explain where each one operates.
- Describe a switch forwarding table and the learn, look up, forward and flood behaviour, including MAC flooding.
- Describe firewalls, intrusion detection systems and intrusion prevention systems, and state what each one does and does not do.
- Explain load balancing algorithms and the difference between a forward proxy and a reverse proxy.
- Compare network attached storage with a storage area network, and name the protocols each uses.
- Define network functions virtualisation, virtual private clouds, security groups, network ACLs and the cloud gateways.
- Compare public, private and hybrid deployment models with software, infrastructure and platform as a service, and say where responsibility stops.
- Compare static, distance vector, link state and path vector routing, including RIP, OSPF, EIGRP and BGP.
- Explain administrative distance, metrics and the longest match rule, and the order in which they decide a route.
- Describe network address translation and port address translation, including the inside, outside, local and global vocabulary.
- Explain first hop redundancy protocols and why the default gateway needs one.
- Explain subinterfaces and router on a stick, and the bottleneck that design introduces.
- Build a VLAN plan, explain 802.1Q tagging, trunk and access ports, and the VLAN hopping attacks that target the native VLAN.

- Explain the switching loop problem, root bridge election, spanning tree port states, and the Rapid and Multiple Spanning Tree variants.
- Describe the spanning tree guard features and port security configuration that protect an access port.
- Explain the maximum transmission unit, jumbo frames and why a mismatch causes partial failures.
- Compare wireless bands, channels, network identifiers and modes, and the encryption history from WEP to WPA3.
- Explain enterprise wireless authentication with 802.1X and EAP, and how a guest network should be designed.
- Describe rack, cabling and environmental practice, and why physical access defeats logical controls.
- Map network segments to security zones, including the DMZ with its external and internal firewalls, and explain microsegmentation.

Course context

Item	Detail
Course	Network Fundamentals + CompTIA Security+ SY0-701, 12 sessions of 3 hours
Strand	Network foundation (sessions 1–3)
Prerequisites	Session 1, networking models and IP addressing
Assessment	HTML learning quiz (20 questions) + Wayground quiz (15 questions)
Exam link	Security+ Domain 3 architecture and Domain 4 operations build on this material

Session 2 coverage

Chapter	Area	What is covered
1	Infrastructure devices	Switches and routers, MAC learning, forwarding table and flooding, firewalls, IDS and IPS, load balancers, proxies, NAS and SAN, wireless hardware
2	Cloud and routing foundations	Network functions virtualisation, virtual private clouds, security groups, service and deployment models, static routing, distance vector and RIP, OSPF
3	Route selection and translation	EIGRP and BGP, administrative distance, metrics, the longest match rule, NAT and PAT, first hop redundancy, router on a stick
4	VLANs, switching and spanning tree	VLAN fundamentals and design, 802.1Q trunking, port configuration and security, loops, root bridge, STP states, RSTP and MSTP, guard features, MTU and jumbo frames
5	Wireless and physical deployment	Channels and bands, SSIDs and modes, WEP to WPA3, 802.1X and EAP, guest networks, access points and antennas, racks, power and environment

How this session maps to the course

Session 8 (enterprise network security) builds directly on the firewall, IDS, IPS and zone material here, and Session 9 (cloud and virtualisation security) picks up network functions virtualisation, security groups and the shared responsibility boundary. Session 5 (threats) references MAC flooding, VLAN hopping and rogue root bridges as attack vectors. Segmentation appears in the exam as an architectural control: it reduces the attack surface, limits blast radius and blocks lateral movement, so revisit this session when you meet zone-based design and microsegmentation later. Every quiz question is tagged with its topic area.